

1. Administrative & Study Identification

**Full Study Title:** A Dose Finding Study of [177Lu]Lu-DOTA-TATE in Newly Diagnosed Glioblastoma in Combination With Standard of Care and in Recurrent Glioblastoma as a Single Agent **Short Title/ Acronym:** Phase 1b [177Lu]Lu-DOTA-TATE in Glioblastoma **Protocol Number:** CAAA601A52101 **NCT Number:** NCT05109728 **Sponsor Name:** Novartis **Investigational Product:** [177Lu]Lu-DOTA-TATE (Targeted Radionuclide Therapy) **Phase of Development:** Phase I (Phase 1b) **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To establish and determine the recommended dose and assess the dose-limiting toxicities of [177Lu]Lu-DOTA-TATE in combination with the standard of care in newly diagnosed glioblastoma, and as a single agent in recurrent glioblastoma. **Secondary Objectives:** To evaluate the overall objective status (via modified RANO criteria), progression-free survival (PFS), overall survival (OS), and overall safety and tolerability.

**2.2 Background & Rationale** Glioblastoma is the most common and aggressive primary brain tumor in adults, characterized by a high mortality rate. Standard of care—maximal safe resection followed by radiotherapy and temozolomide—yields limited long-term survival. This study introduces targeted radionuclide therapy (RLT) utilizing [177Lu]Lu-DOTA-TATE to target the Somatostatin Receptor 2 (SSTR2), which is being explored as a novel mechanism to deliver localized radiation directly to neoplastic cells and improve clinical outcomes.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Single-arm (Dose-finding) **Blinding:** Open-label **Randomization:** N/A

**3.2 Planned Enrollment & Duration Target \$N\$:** 60 participants **Number of Sites:** Global, multicenter (including U.S. sites such as University of Wisconsin, University of Pittsburgh Cancer Institute, and MD Anderson Cancer Center) **Estimated Study Start:** Late 2021 **Estimated Primary Completion:** 30-Jul-2026

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Participant is  $\geq 18$  years of age on the day of signing the informed consent form. 2. Histologically confirmed glioblastoma (Newly diagnosed with KPS  $\geq 70\%$  for Group 1; Recurrent with KPS  $\geq 60\%$  for Group 3). 3. Adequate bone marrow, organ function, and electrolyte values.

**4.2 Exclusion Criteria** 1. Participant is receiving additional, concurrent, active therapy for glioblastoma outside of the trial. 2. Extensive leptomeningeal disease. 3. History of another active malignancy in the previous 3 years prior to study entry or prior administration of a radiopharmaceutical unless 10 or more effective half-lives have elapsed.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** [177Lu]Lu-DOTA-TATE administered intravenously every 4 weeks (for Group 1 newly diagnosed) or every 3 weeks  $\pm 2$  days (for Group 3 recurrent glioblastoma). **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Up to 6 administrations for Group 1 (approx. 24 weeks). Followed by a 12-month follow-up period.

**5.2 Concomitant & Prohibited Therapy Permitted:** Radiotherapy (60 Gy) and temozolomide (concomitant 75 mg/m<sup>2</sup>/day; maintenance escalating up to 200 mg/m<sup>2</sup>) for Group 1. Sterile 2.5% Lysine-Arginine amino acid solution is co-administered for renal protection. **Prohibited:** Additional active therapy for glioblastoma outside of the trial.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days)            | Key Activities/Procedures                                                                                          |
|----------|--------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1        | Screening                | Up to 6 weeks   Informed Consent, [68Ga]Ga-DOTA-TATE PET scan, baseline safety labs                                |
| 2        | Baseline                 | Day 0   First Dose, safety assessments                                                                             |
| 3-7      | Interim                  | Every 3 to 4 Weeks   [177Lu]Lu-DOTA-TATE infusion, amino acid infusion, standard of care therapies (as applicable) |
| 8-12     | End of Study / Follow-up | Up to 12 months   Follow-up visits every 8 weeks for MRI/tumor assessments and survival tracking                   |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Incidence of dose-limiting toxicities (DLTs) to establish the recommended dose. **Measurement Method:** Clinical observation, safety lab parameters, and toxicity grading reviewed by the dose-escalation committee.

**7.2 Secondary & Exploratory Endpoints** 1. Best objective overall status (Confirmed or Preliminary Complete Response/Partial Response) according to modified RANO criteria. 2. Progression-Free Survival (PFS) and Overall Survival (OS).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous clinical and laboratory evaluation throughout the treatment and follow-up phases. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor. **Data Monitoring Committee (DMC):** An internal Dose Escalation/Safety Review Committee will evaluate DLTs and safety data from each cohort prior to escalating the dosage or expanding the groups.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Full Analysis Set (FAS) for efficacy endpoints (like PFS); Safety Set for all toxicity and adverse event summaries. **Sample Size Justification:** Approximately 60 participants. Justified by Phase 1b standard dose-escalation statistical rules to accurately identify dose-limiting toxicities. **Handling of Missing Data:** For time-to-event endpoints (PFS, OS), Kaplan-Meier methods will be utilized. If no PFS event is observed, data will be censored at the date of the last adequate tumor assessment prior to the cut-off date or start of new anti-neoplastic therapy.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard phase I clinical monitoring involving regular onsite and remote evaluations. **Audit Trail:** Robust Electronic Case Report Form (eCRF) tracking with independent central review for complex imaging (PET/MRI) scans.

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## 11. Study Identifiers & Governance

**Primary ID:** CAAA601A52101 **Registry Number:** NCT05109728 **Sponsor/Lead Organization:** Novartis **Study Title:** A Dose Finding Study of [177Lu]Lu-DOTA-TATE in Newly Diagnosed Glioblastoma in Combination With Standard of Care and in Recurrent Glioblastoma as a Single Agent **Official Regulatory Title:** A Dose Finding Study of [177Lu]Lu-DOTA-TATE in Newly Diagnosed Glioblastoma in Combination With Standard of Care and in Recurrent Glioblastoma as a Single Agent

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## 12. Clinical Framework (Glioblastoma Specific)

**Primary Condition:** Glioblastoma **Diagnosis Threshold:** Histologically confirmed Glioblastoma (Newly Diagnosed or Recurrent) **Medical Methodology:** Targeted Radionuclide Therapy (RLT) **Optimization Target:** Somatostatin Receptor 2 (SSTR2) targeted dose finding

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard of Care (Radiotherapy + Temozolomide) + [177Lu]Lu-DOTA-TATE, or [177Lu]Lu-DOTA-TATE single agent **Education Protocol (HCP):** Site initiation visits and radioligand safety protocol training **Education Protocol (Patient):** Informed consent, radiation safety counseling, and standard neuro-oncology support **Quality Audit Mechanism:** Clinical and safety assessments tracking via sponsor monitoring

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## 14. Observation & Follow-Up Schedule

**Total Duration:** 12 months follow-up **Onsite Visit Cadence:** Every 3-4 weeks during the treatment period **Remote Monitoring Frequency:** Continuous safety and survival monitoring **Checklist Review Interval:** Every 8 weeks (MRI/tumor assessment)

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## 15. Sign-off & Approval

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|--|--------------------------|--|------|---------------|-----------|--|-------|--|--|---------------|--|------|--|------|--|------|--|
|  | Role                     |  | Name |               | Signature |  | Date  |  |  | :---          |  | :--- |  | :--- |  | :--- |  |
|  | Principal Investigator   |  |      | [Insert Name] |           |  | _____ |  |  | [DD-MMM-YYYY] |  |      |  |      |  |      |  |
|  | Clinical Operations Lead |  |      | [Insert Name] |           |  | _____ |  |  | [DD-MMM-YYYY] |  |      |  |      |  |      |  |
|  | Lead Statistician        |  |      | [Insert Name] |           |  | _____ |  |  | [DD-MMM-YYYY] |  |      |  |      |  |      |  |